

Module 1: Modeling premixed laminar flames with chem1d

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Installing and running chem1d

1. Download the zipped folder CHEM1D.zip from CANVAS to your computer and unzip it. This folder contains the installer chem1d-3.2.2.exe and several input and output files of chem1d. The input files settings.csf and defaults.csf are text files containing keywords and values, which are used to control the code.
2. Install chem1d by running the installer. Add chem1d to the PATH for all users by selecting the radio button.
3. Open a command window (cmd.exe from the start menu) and cd to your copy of the CHEM1D folder (not the folder where chem1d.exe has been installed). Run the code by typing chem1d. The screen output of the code is also written in the file chem1d.log. In this file you can check whether the solver has found a converged solution. Some results (burning velocity, temperature, etc.) are listed at the end of this file.
4. The solution (species mass fractions, temperature, velocity, etc. as functions of position x) is stored in the file yiend.dat. The solution can be visualized by using the MATLAB script plotexample.m.
5. In order to run the code with different settings, you have to modify the case files. You can open both case files (.csf) using a plain text editor, e.g., Notepad++, WordPad (set default program, set "no wrap") or the editor of MATLAB. The code first reads the file defaults.csf and subsequently the settings from settings.csf are read, overwriting the defaults settings.

1 Premixed methane-air

1.1 Stoichiometry

Compute a stoichiometric premixed flame using the initial unchanged settings. Take a look at the temperature and species profiles and try to understand/explain them. Is the computational domain large enough? Are you using enough grid points?

Now change the stoichiometry φ of the flame by changing [BOUNDARY_EQUIVALENCERATIO] in settings.csf from 0.95 to 0.90. Leave the second entry on the line unchanged (it is not used anyway). Save the settings and run chem1d again. Take a look at the screen output and analyze the results (especially burning velocity s_L and flame temperature T_b). Continue this procedure and compute flames for $\varphi = 0.6, \dots, 1.3$. If the solver does not converge within, say, 1000 iterations, stop the code by pressing Ctrl+C and apply a smaller step in φ . Store the yiend.dat solutions for later use by copying them into a folder using logical names, e.g.,

output/yi_0.950.dat.

If you are lazy, you can use the matlab script runseries.m, which performs this procedure automatically.

Plot s_L and T_b as function of φ . The flame temperature T_b is the temperature of the burnt gas and it can be taken at the last grid point. Make also a plot of CO and NO emissions versus equivalence ratio φ . The CO and NO emissions may be represented by the mass fractions of CO and NO at the last grid point of the domain.

1.2 Simplified diffusion models

In chem1d, diffusion of species is by default modeled by using mixture-averaged diffusion coefficients. We want to know what the effect is of assuming constant Lewis numbers. In order to do that we first have to determine the Lewis numbers. This can be done by chem1d. It fits constant values to the mixture-averaged coefficients, which vary with x . Run chem1d with the output option LEWIS at [POSTPROCESSING_OUTPUTFILES] in defaults.csf turned ON. After the run a file lewis.dat will be written which contains the Lewis numbers. Check this! What is the Lewis number of methane and which species has the lowest Lewis number?

Now we can use these Lewis numbers by changing the transport model option [MODEL_TRANSPORT] in defaults.csf to LEWIS. Make sure that the name of the Lewis number file is set correctly at [LEWIS_TRANSPORTFILE]. Run the same loop over φ as above and determine the burning velocity and flame temperature as function of φ . Compare the results with the mixture-averaged model.

In many theoretical and numerical models, unity Lewis numbers are assumed ($Le_i = 1$ for all i). Investigate what the effect of this simplification is on burning velocity and temperature. The Lewis numbers can be set to one for certain species by simply removing them from the Lewis number file. If chem1d doesn't find a species name in that file, it assumes a unity Lewis number for that species. So, if you create an empty Lewis number file, the Lewis numbers of all species will be set to one.

Again, determine the laminar burning velocity s_L and flame temperature T_b as function of φ and compare with the other two diffusion models.

1.3 Temperature and pressure dependence

Revert to the original settings: $\varphi = 0.9$ and mixture-averaged diffusion coefficients. Investigate the dependence of burning velocity on temperature and pressure. The pressure can be varied using the option [BOUNDARY INLETPRESSURE]. Change the pressure from 1 to 10 bar in small steps. Plot s_L/s_L^0 versus p/p^0 , where the superscript 0 denotes the reference value at standard conditions. How does the burning speed scale with pressure? What about the mass burning rate $\rho_u s_L = (\rho_u s_L)^0$?

Do the same for the inlet temperature T_u . Increase it from 300 to 500 K by changing the option [BOUNDARY INLETTEMPERATURE]. Note that this options needs two values: the first one corresponds to the fuel, the second to the oxidizer stream. Plot $\ln(\rho_u s_L)$ as a function of $1/T_b$ and compute the activation energy E_a from the slope of this curve.

Another method to change the flame temperature T_b is to change the amount of (nitrogen) dilution. If you decrease the amount of N₂ in the oxidizer, the flame temperature will increase and vice versa. Try this and compare with the variations in T_u .

Do you find the same s_L for the same T_b ? What about the mass burning rate $m = \rho_u s_L$?

1.4 Burner-stabilized flat flames

To investigate the stabilization of flat flames, we enter the directory burner-stabilized and run the default case. This is a lean ($\varphi=0.9$) premixed methane-air burner-stabilized flame with an inlet velocity $u_u = 31$ cm/s (i.e. mass flux $m = \rho_u u_u = 0.035$ g/(cm s²)). Note that the flame temperature T_b is lower than the adiabatic one due to heat loss.

Lower the mass flow in small steps by changing the option [BOUNDARY MASSFLOW] in the settings file. Run a series of calculations with $m = 0.010, \dots, 0.035$ g/(cm s²). Plot again $\ln(\rho_u s_L)$ as a function of $1/T_b$ and compare with your results of the previous section.